Anomaly Detection in Industrial Images Using Neural Networks

Team 14 - BA865

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Prediction Task & Motivation

Prediction Task

The objective of our project is to build a neural network model capable of detecting anomalies in industrial product images.

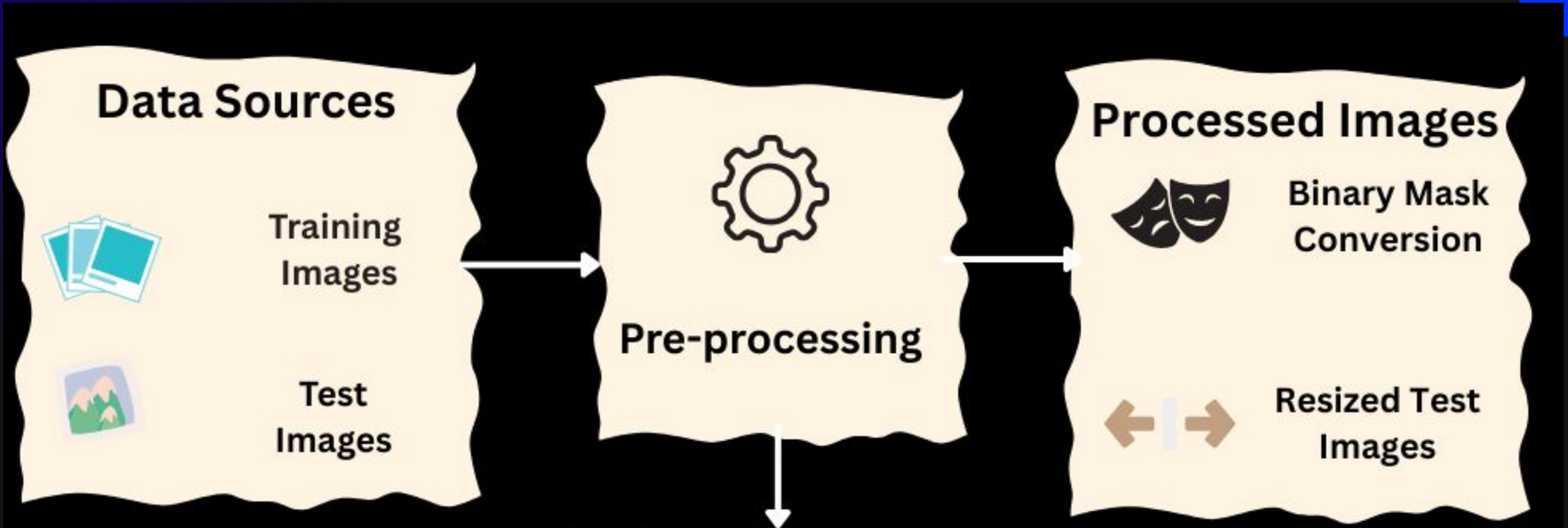
We train a convolutional autoencoder to learn the structure of defect-free (normal) images from the MVTec AD dataset.

The model trains and learns to identify the defects in the image object. As a next step, we want it to classify specific defect types (e.g., scratch, contamination, missing parts) of the object as well.

Motivation

- Manual inspection is costly and inconsistent.
- Automated detection = faster, cheaper, more reliable.
- Helps in scaling industrial production.
- Supports smarter, AI-driven quality control.

Pre-processing



Dataset Structure

- Loaded from MVTEC AD opuset (carpet, transistor, hazeinut)
- Spilt into
- Train - defect-free images only
- Ground Truth - pixel-level binary defect masixs

Image Resizing

- RGB images reized 128x128.pix
- Reducing dimensionality and noise

- **Binary Mask Conversion**

Converted ground truth t60, 1 [img/ 255.0]

- **Normalization**

Scaled pixel value [O, 1]

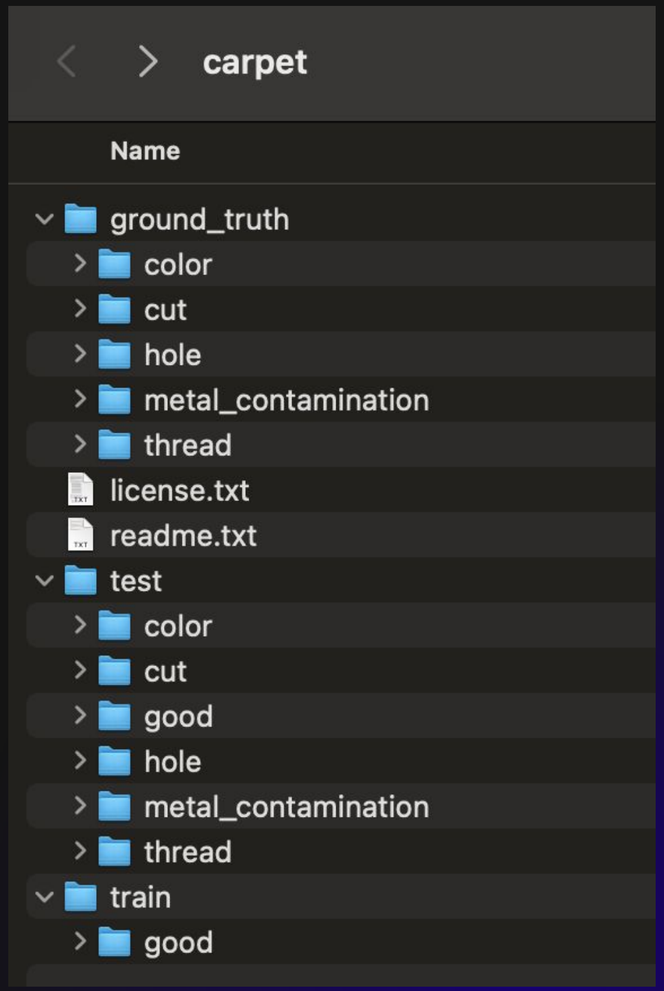
- **Binary Mask Thresholding**

Data Dictionary

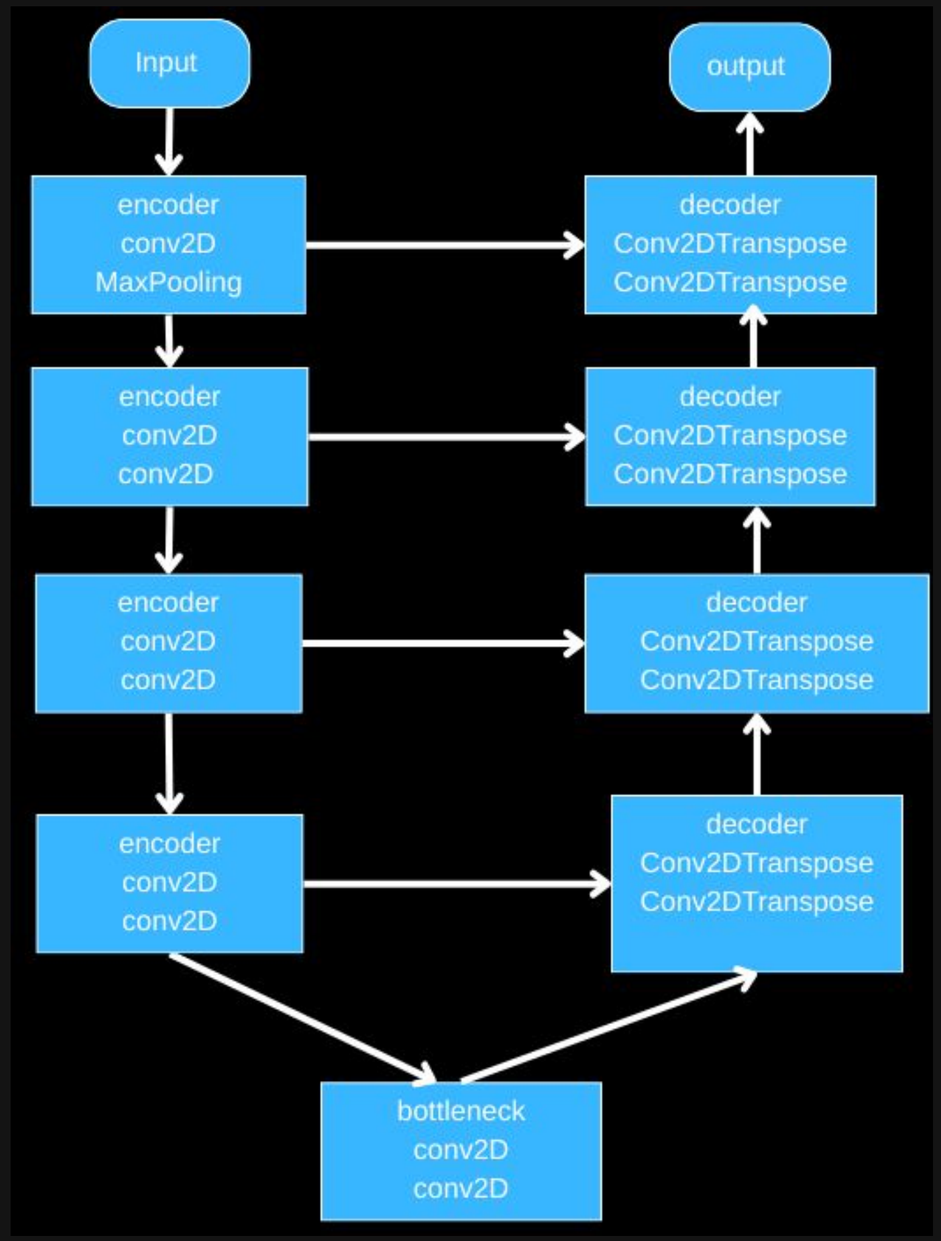
- 'train', which contains the (defect-free) training images
- 'test', which contains the test images
- 'ground_truth', which contains the pixel-precise annotations of anomalous regions

Our dataset is organized into three main parts:

- **Train** — This folder contains only normal (defect-free) images. These are used to teach the model what "healthy" data looks like.
- **Test** — This folder includes both good and defective images. The defective ones are sorted into categories like *cut*, *hole*, *metal contamination*, etc.
- **Ground Truth** — For each defective test image, this folder provides a pixel-level mask that shows exactly where the anomaly is.



Model Architecture



Conv2D Layers: Extract local spatial features while reducing noise and irrelevant details.

MaxPooling: Downsamples feature maps to reduce dimensionality and highlight dominant patterns.

Bottleneck Layer: Captures the most compressed and essential representation of the input.

Conv2DTranspose: Reconstructs the input resolution by upsampling learned features.

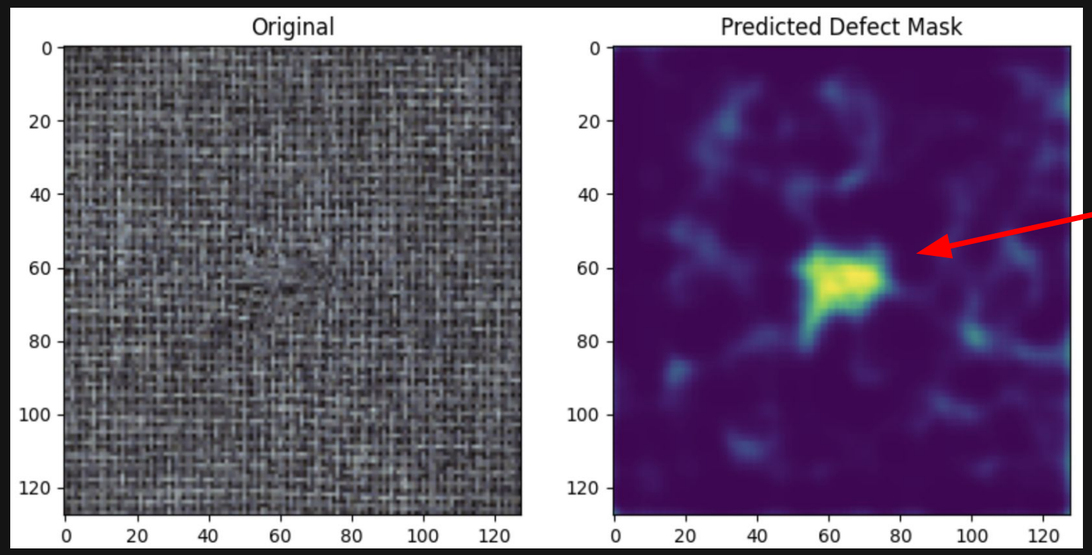
Skip Connections: Preserve spatial detail by linking encoder and decoder layers, enhancing reconstruction accuracy.

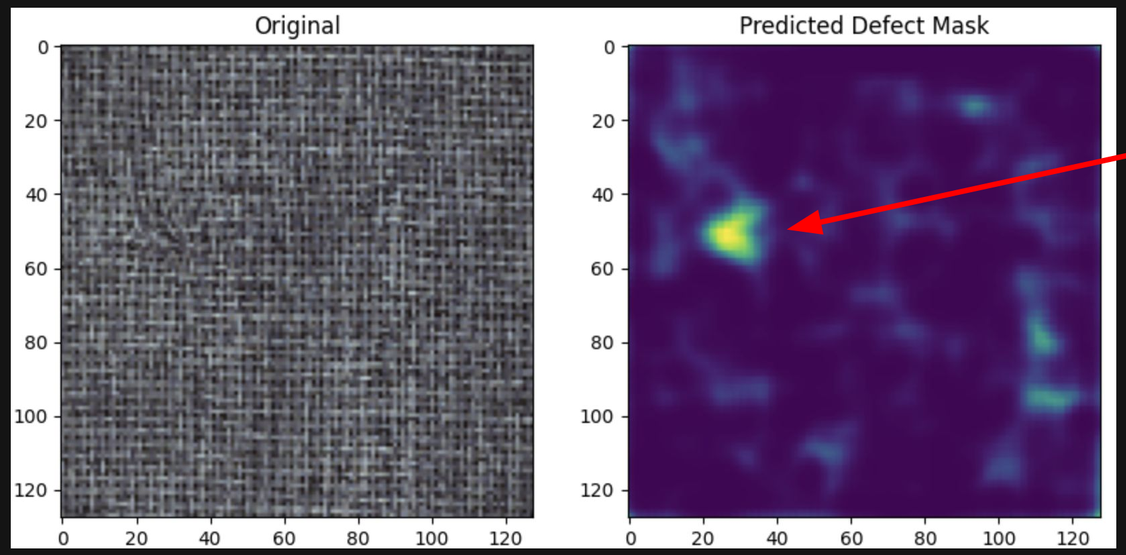
Activation Function: Sigmoid is used in the final output layer to produce pixel-wise probabilities between 0 and 1 — ideal for binary segmentation masks.

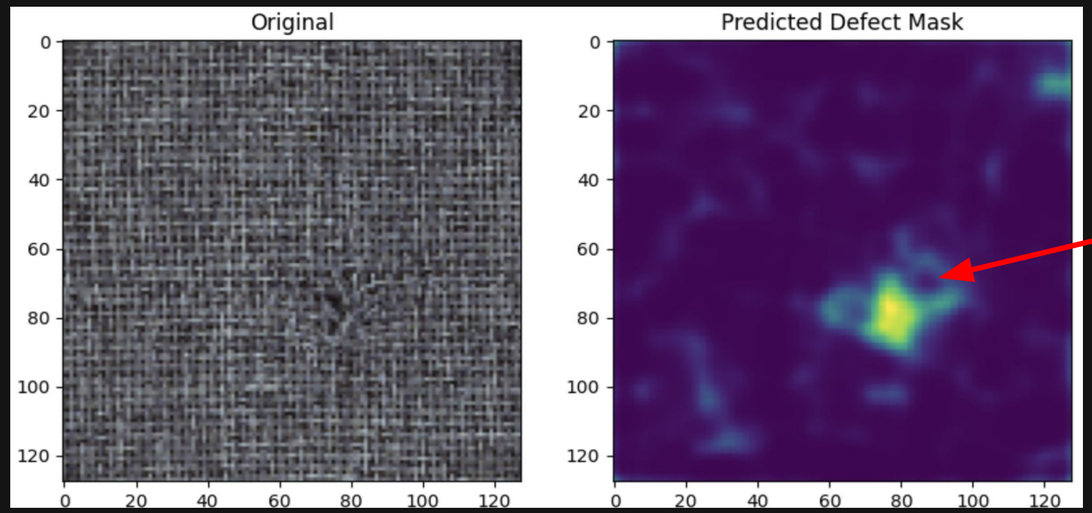
Loss Function: Adam optimizer is used for its adaptive learning rate and fast convergence, making training efficient even on limited data.

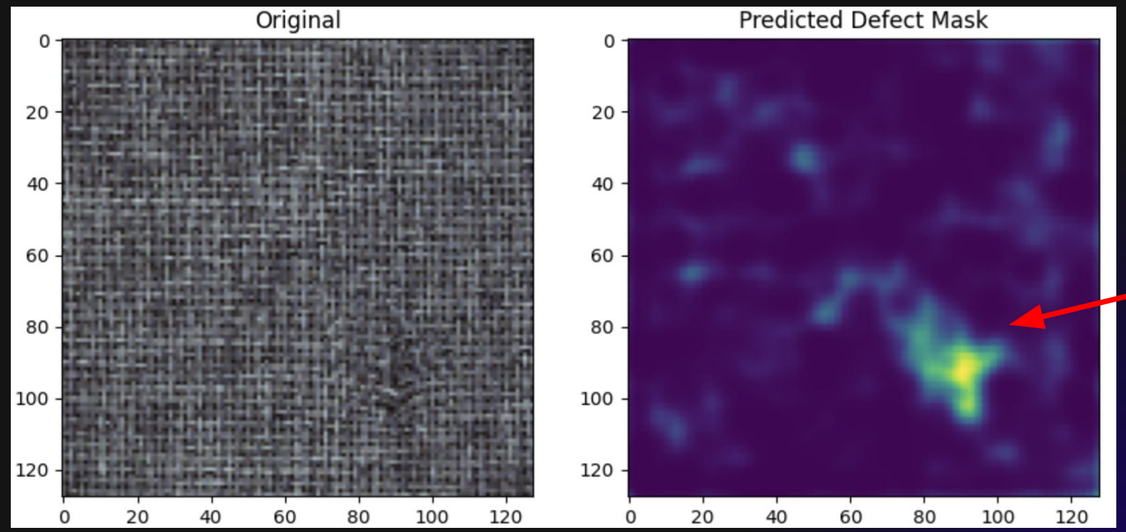
Optimizer: Binary Crossentropy is used to measure the difference between predicted and actual binary masks, ideal for pixel-level anomaly detection.

Results - Carpet

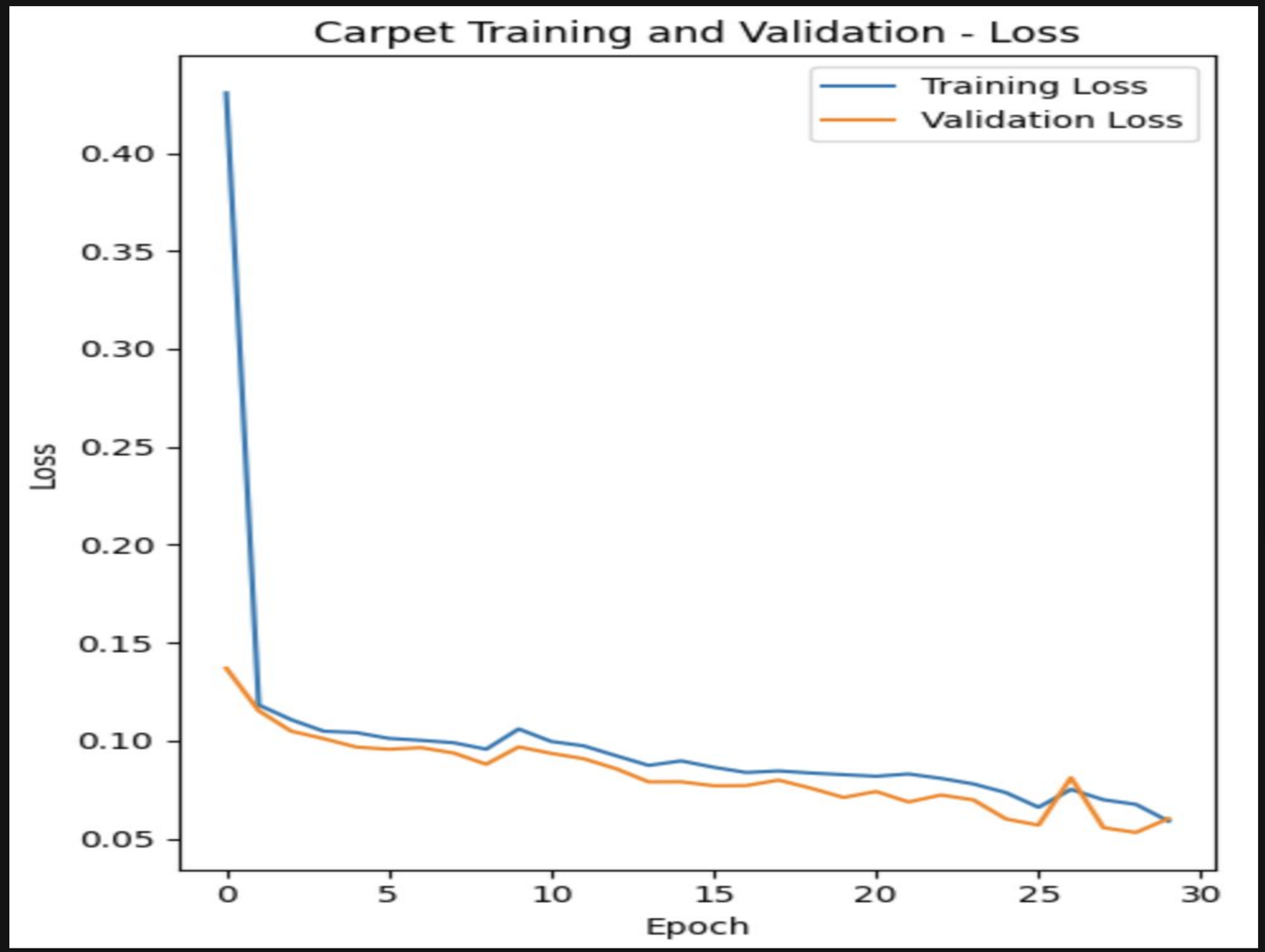






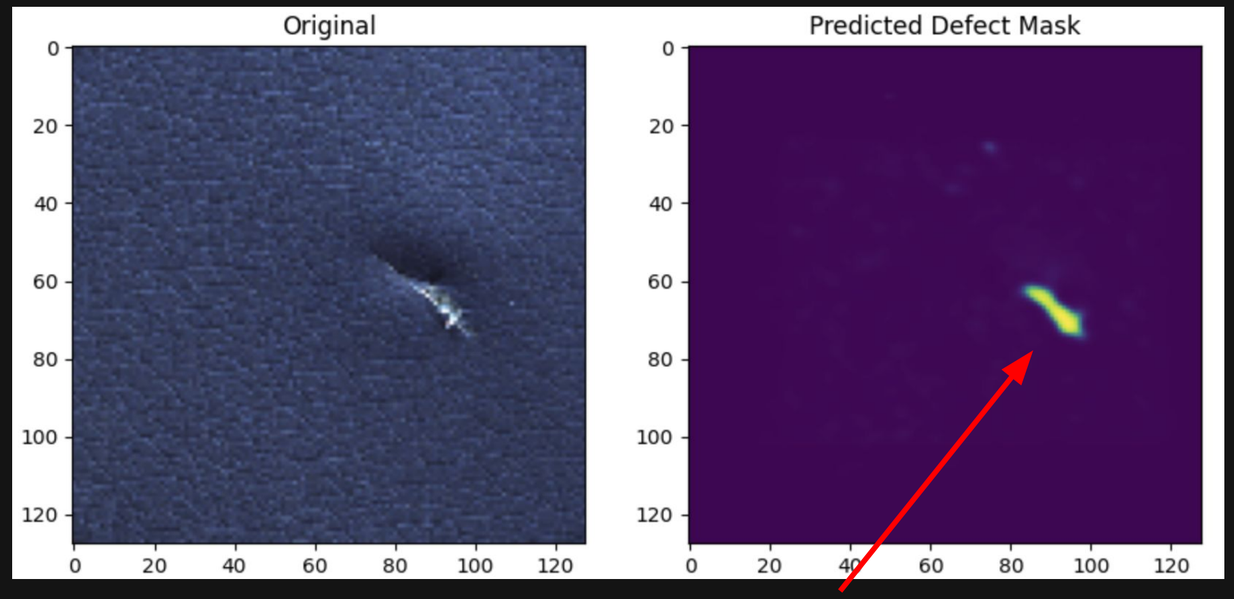


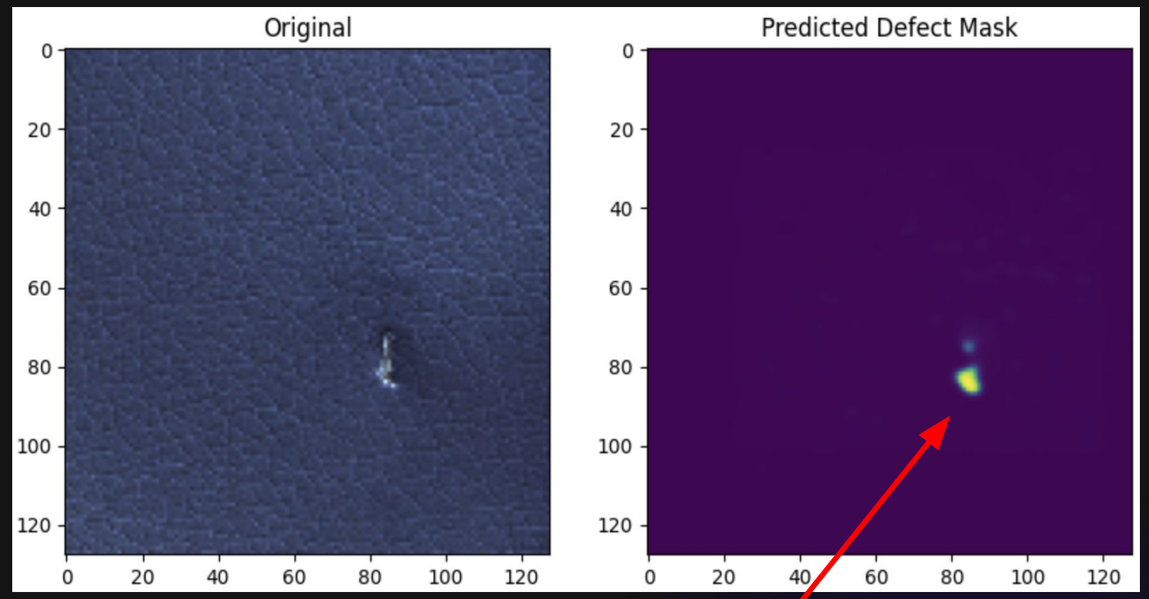
Loss - Carpet

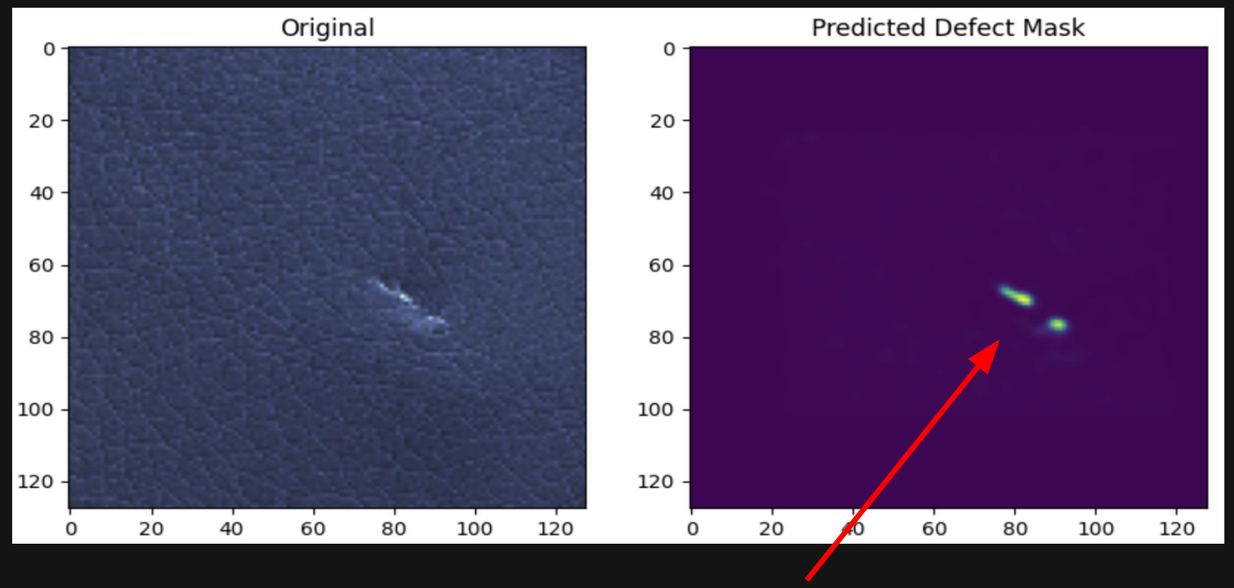


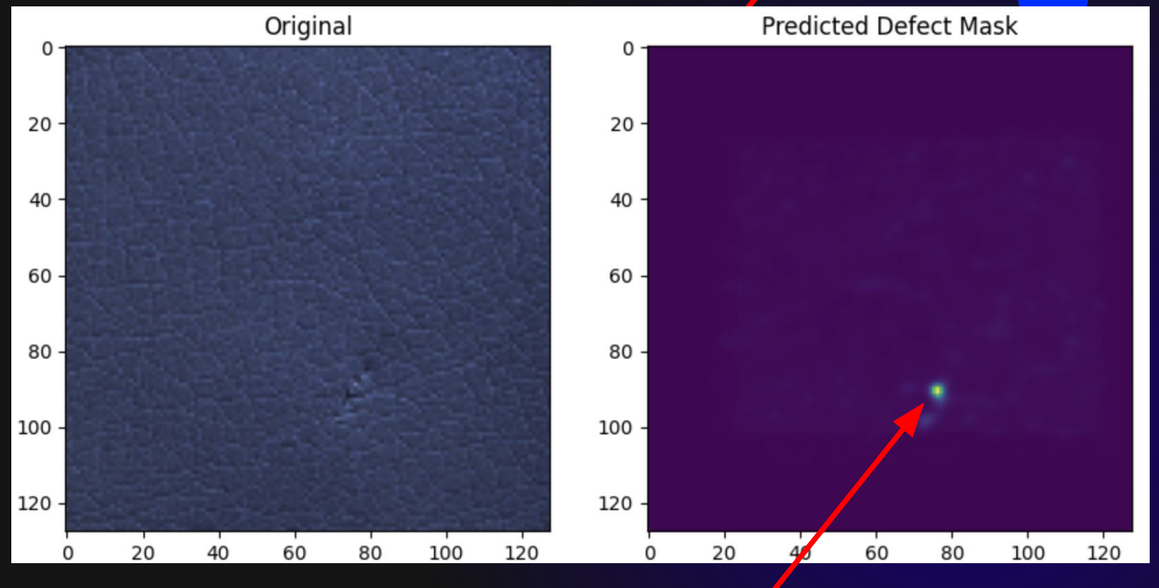
- Fast convergence: Both training and validation losses drop sharply in the first 3–5 epochs, quickly reaching below 0.12, signaling effective learning.
- Consistent performance: From epoch 10 onward, loss values gradually decrease and stabilize around 0.06–0.08.
- Strong generalization: The close alignment between training and validation loss curves throughout the epochs confirms minimal overfitting.
- Smooth progression: Only minor fluctuations occur, suggesting the model is reliably learning patterns in carpet defect data.

Results - Leather

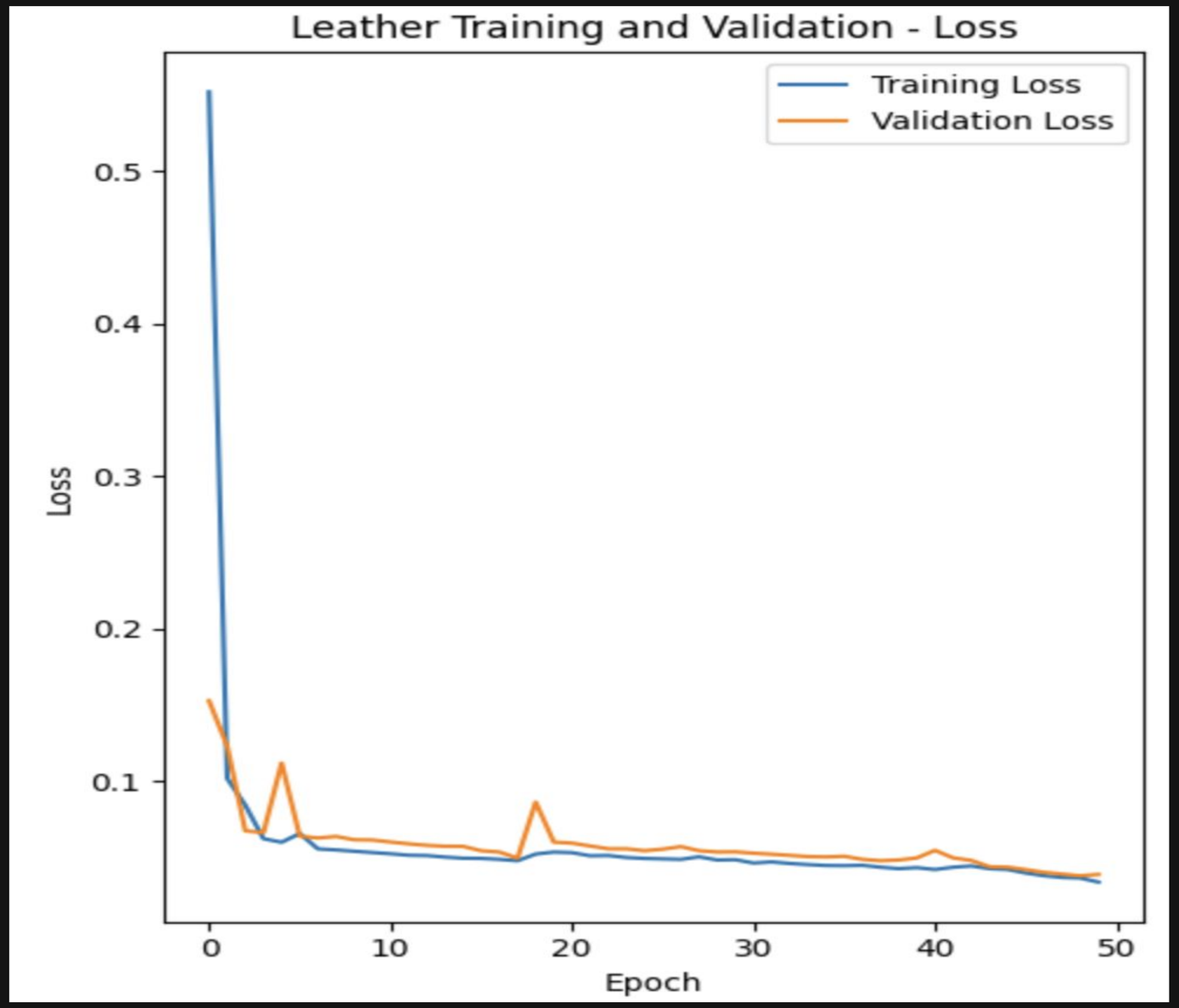








Loss - Leather



- The model achieves rapid convergence within the first 5 epochs, with both training and validation loss dropping below 0.1, indicating efficient learning.
- After an initial sharp drop, both losses stabilize around 0.04–0.06, showing that the model captures the defect features effectively.
- Minor fluctuations in validation loss are observed around epochs 5–20, but no signs of overfitting are present.
- The model maintains consistent performance across the full 50 epochs, suggesting strong generalization to unseen leather defect data.

Challenges:

- Transistor components are visually complex and difficult to reconstruct.
- Bright reflections or lighting variations often trigger false positives.
- Autoencoder outputs are overly smooth, missing fine details in defects.

Limitations:

- Model detects defective areas but doesn't label defect types yet.
- Processing is computationally heavy due to high-resolution images.
- Thresholding is static and doesn't adapt to defect variability.

Conclusion:

- A U-Net model was trained on defect-free leather images to detect and segment surface anomalies.
- The model achieved low and stable training and validation losses, indicating effective learning and minimal overfitting.
- It successfully identified various defect types (e.g., scratches, contamination) with clear and accurate segmentation masks.
- The model generalized well across test data, demonstrating consistency and reliability for defect detection tasks.
- Overall, the system shows strong potential for automating visual inspection in industrial quality control.

Future Steps:

- Apply advanced data augmentation techniques (rotation, noise, elastic transform) to make the model more robust against real-world variations.
- Expand to unsupervised anomaly detection using Variational Autoencoders (VAE) to detect defects without manual labels.
- Apply SHAP or LIME to enhance model explainability, providing visual insight into defect detection decisions.

Thank You